ModelIt!

Teacher's Guide

How Do Doctors See Inside You Without Cutting You Open?

Modeling MRI Physics and Medical Imaging Technology

Quick Reference

Grade Level	Grade 10L2
NGSS Standard	HS-PS2-4, HS-PS4-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS2-4, HS-PS4-4

Use mathematical representations of Newton's law of gravitation and Coulomb's law to describe and predict the gravitational and electrostatic forces between objects. Evaluate the validity and reliability of claims in published materials of the effects that different frequencies of electromagnetic radiation have when absorbed by matter.

Three-Dimensional Learning

Science Practice	**Evaluating Claims** — Students evaluate claims about medical imaging technologies by modeling the physics-based trade-offs between field strength, image quality, safety, and cost.
Disciplinary Core Idea	**PS2.B Types of Interactions / PS4.B Electromagnetic Radiation** — Magnetic fields exert forces on aligned nuclear magnetic moments; radiofrequency electromagnetic radiation can be absorbed and re-emitted by atomic nuclei, with tissue-dependent properties that enable medical imaging.
Crosscutting Concept	**Structure and Function** — Students analyze how the physical structure of MRI components (superconducting coils, gradient coils, RF transmitters) determines the functional capability of the imaging system.

CAST Testing Alignment

MULTIPLE CHOICE:

A hospital is choosing between a 1.5T MRI and a 3T MRI. Based on your model, which factor would most improve by upgrading to 3T?

CONSTRUCTED RESPONSE:

Using your model, explain how MRI creates images of soft tissues without using X-rays or any ionizing radiation. Reference Magnetic Field Strength, Tissue Contrast, and the role of hydrogen nuclei in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Magnetic Resonance Imaging is one of the most remarkable achievements of applied physics — it turns the quantum mechanical properties of hydrogen atoms into detailed images of the human body's interior. Without a single X-ray photon or surgical incision, MRI can distinguish between healthy and diseased tissue, map brain activity, visualize blood flow, and detect tumors as small as a few millimeters.

The Physics of Nuclear Magnetic Resonance

Every hydrogen atom has a nucleus (single proton) that acts as a tiny magnetic dipole — a miniature bar magnet. Normally, these nuclear magnets point in random directions, canceling out. But when placed in a strong external magnetic field, the protons align either parallel or antiparallel to the field, with a slight majority parallel (lower energy state). This creates a net magnetization that can be manipulated with radio waves. When a radiofrequency pulse at the precise resonance frequency (the Larmor frequency, proportional to field strength) is applied, it flips the aligned protons to the higher energy state. When the pulse stops, the protons relax back to equilibrium, emitting radiofrequency signals as they do. These signals are the raw data of MRI.

How Contrast Works

The key to MRI's diagnostic power is that different tissues have different relaxation times. T1 relaxation (spin-lattice) measures how quickly protons return to alignment with the main field. T2 relaxation (spin-spin) measures how quickly the precessing protons fall out of phase with each other. Fat has short T1 (recovers quickly, appears bright on T1-weighted images). Cerebrospinal fluid has long T1 and long T2 (appears dark on T1, bright on T2). Tumors often have different relaxation times from surrounding healthy tissue, making them visible. By adjusting the timing of RF pulses (the pulse sequence), radiologists can create images that highlight specific tissue properties.

The Superconducting Magnet

Creating a stable, powerful, and homogeneous magnetic field requires a superconducting electromagnet. Miles of niobium-titanium wire are wound into coils and cooled to 4.2 Kelvin (-269 degrees Celsius) using liquid helium. At this temperature, the wire's electrical resistance drops to exactly zero, allowing thousands of amperes to flow without energy loss. Once energized, the current circulates indefinitely — many MRI magnets have been running continuously for years without being re-energized. But the system is fragile: if any section of the coil warms above the critical temperature, resistance appears, generating heat that boils the helium explosively in a 'quench' — a dramatic and expensive failure.

Safety and Limitations

MRI is generally very safe because it uses non-ionizing radiation (radio waves, not X-rays). However, the powerful magnetic field creates unique hazards. Ferromagnetic objects (iron, steel) become high-velocity projectiles when brought near the scanner — MRI-related accidents from chairs, oxygen tanks, and tools being pulled into the bore have caused injuries and deaths. Patients with certain metallic implants (pacemakers, cochlear implants, shrapnel) cannot safely enter the magnetic field. The strong radiofrequency pulses deposit energy in tissue (measured as Specific Absorption Rate, SAR), which can cause heating. The loud noise from gradient coil switching (up to 130 dB) requires hearing protection.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify seven components of the MRI system: Magnetic Field Strength, Coil Temperature, Image Resolution, Tissue Contrast, Patient Safety, Energy Consumption, and Diagnostic Accuracy — with Magnetic Field Strength and Coil Temperature as external engineering parameters.

E **Establish Relationships**

Students determine that Magnetic Field Strength drives Image Resolution and Tissue Contrast (both improving signal-to-noise ratio), which together determine Diagnostic Accuracy. Higher field strength increases Energy Consumption and creates greater Patient Safety concerns. Coil Temperature must stay below the superconducting critical point.

V **Visualize & Model**

Students build a computational model showing how field strength affects all downstream parameters and the trade-offs between image quality, safety, and cost.

E **Evaluate Outcomes**

Students run 1.5T standard, 3T high-field, and coil failure scenarios to compare image quality, safety profiles, and the consequences of system failure.

R **Revise & Extend**

Students add Gradient Coil Precision, Scan Duration, or RF Pulse Sequence Design to explore how additional engineering parameters affect MRI performance optimization.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This machine can see inside your skull without touching you. No X-rays. No cutting. Just magnets and radio waves. How is that even possible?"*

Watch For

Show a stunning MRI brain scan image. Ask: How can a magnet create a picture of something it can't touch? Collect initial hypotheses.

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling one of the most sophisticated machines in modern medicine — and the physics that makes it work is something you already know: electromagnetism."*

Watch For

Have students read objectives. Pre-teach 'nuclear magnetic resonance' and 'relaxation time.' Quick-write: What do you think happens to atoms in a magnetic field?

Slide 3: Big Question

3 min

Say: *"Your body is 60% water. Water is H2O. Those hydrogen atoms are about to become the most important atoms in medicine."*

Watch For

Students estimate: How many hydrogen atoms are in your body? (Answer: approximately 4 x 10^27.) Ask: What if you could make every single one of them send you a signal?

Slide 4: LEVER Framework

2 min

Say: *"We're going to model how magnets, cold, and radio waves combine to create the most detailed images of the human body ever possible."*

Watch For

Preview each LEVER step. Emphasize that MRI connects quantum physics (nuclear spin) to clinical medicine (diagnosis).

Slide 5: Activity 1: Components

8 min

Say: *"Two things engineers control: how strong to make the magnet and how cold to keep it. The physics and the patient determine everything else."*

Watch For

Guide component sorting. Discuss why Magnetic Field Strength and Coil Temperature are external (engineering choices) while image quality and safety are internal responses.

Slide 6: Activity 2: Connections

8 min

Say: *"Stronger magnet means better images. But it also means more energy, more danger, and one malfunction away from a catastrophic quench."*

Watch For

Help students map the trade-offs: field strength improves image quality but increases energy, cost, and safety concerns. Trace the coil temperature failure cascade.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare a standard hospital MRI, a research powerhouse, and what happens when the cooling fails spectacularly."*

Watch For

Students predict outcomes for each scenario. Run simulations. The quench scenario is particularly dramatic — discuss what happens physically.

Slide 8: Discoveries

5 min

Say: *"You just modeled a $2 million machine that uses the coldest temperature in the hospital to detect the warmest thing in medicine: living tissue."*

Watch For

Lead discussion connecting model findings to real MRI capabilities and limitations. Show clinical MRI images comparing healthy and diseased tissue.

Slide 9: STEM Challenge

12 min

Say: *"70% of the world has no access to MRI. Design the machine that changes that — even if it means sacrificing the Rolls Royce version for something that actually reaches people."*

Watch For

Groups design portable MRI concepts. Must specify magnet type, field strength, target diagnoses, power requirements, and cost estimate. Present as startup pitches.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Magnetic Field Strength, Coil Temperature
- **INTERNAL:** Image Resolution, Tissue Contrast, Patient Safety, Energy Consumption, Diagnostic Accuracy

Relationships

- **Magnetic Field Strength to Image Resolution:** POSITIVE (+) — Higher Magnetic Field Strength increases the signal-to-noise ratio of the detected radiofrequency emissions from hydrogen nuclei. More signal relative to noise allows finer spatial detail to be resolved, improving Image Resolution approximately proportionally to field strength.
- **Magnetic Field Strength to Tissue Contrast:** POSITIVE (+) — Higher Magnetic Field Strength increases the difference in signal intensity between different tissue types because the Boltzmann distribution creates a larger population difference between aligned and anti-aligned spin states, enhancing the contrast-generating relaxation time differences.
- **Image Resolution to Diagnostic Accuracy:** POSITIVE (+) — Higher Image Resolution allows radiologists to detect smaller abnormalities and distinguish fine structural details, directly improving the probability of correct diagnosis. A 1mm resolution MRI can detect tumors invisible on a 3mm resolution scan.

Key Discoveries

- MRI creates images using only magnetic fields and radio waves — no ionizing radiation is involved, making it fundamentally safer than X-rays or CT scans for repeated imaging
- Image Resolution and Tissue Contrast both improve with higher Magnetic Field Strength, but the relationship is not linear — doubling field strength from 1.5T to 3T roughly doubles signal-to-noise ratio
- The superconducting coil is the most critical and fragile component — if Coil Temperature rises above the critical threshold, the magnet 'quenches,' explosively boiling the liquid helium and destroying the magnetic field
- The trade-off between image quality and patient safety means that the most powerful MRI machines are not always the best choice — 1.5T is sufficient for most diagnoses and presents fewer safety challenges

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"MRI uses radiation like X-rays"	MRI uses radiofrequency electromagnetic radiation, which is non-ionizing — the same fundamental type as FM radio signals and Wi-Fi, just at specific frequencies that resonate with hydrogen nuclei. Unlike X-rays and CT scans, MRI does not use ionizing radiation that can damage DNA. This is why MRI is preferred for repeated imaging and for patients where radiation exposure is a concern (children, pregnant women).
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Stronger magnets are always better for MRI"	While higher field strength improves signal-to-noise ratio and image resolution, it also increases safety risks (stronger forces on metallic objects, higher SAR), cost, energy consumption, and technical complexity. For most clinical diagnoses, 1.5T provides sufficient image quality. The trend toward higher fields (3T, 7T) is driven by research applications that need extremely fine detail, not by clinical necessity.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"MRI is perfectly safe for everyone"	While MRI does not use ionizing radiation, it is not risk-free. The powerful magnetic field can turn ferromagnetic objects into dangerous projectiles, cause implanted devices (pacemakers, neurostimulators) to malfunction, and move metallic foreign bodies within the body. The RF pulses deposit thermal energy in tissue, which can be problematic at high field strengths. The gradient coil switching produces noise levels up to 130 dB — loud enough to cause hearing damage without ear protection.